

HEMANT RAJ SEN

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SUMMARY

CSE Student at JUET with hands-on ML implementations built from scratch — gradient descent optimizer, image processing toolkit, and live F1 telemetry analysis. Proficient in Python and data analysis; comfortable moving from mathematical theory to working code. Seeking a summer internship to contribute to real-world ML or data engineering projects.

TECHNICAL SKILLS

Languages	Python (primary), C/C++, SQL, HTML/CSS
ML / Data	NumPy, pandas, matplotlib, scikit-learn, FastF1
CS Fundamentals	DSA, OOP, DBMS, mathematical optimization
Tools	Git, GitHub, Jupyter Notebook, VS Code, Vercel

PROJECTS

Gradient Descent — From Scratch | *Python, NumPy, Mathematical Modeling* [GitHub ↗](#)

- Implemented the full gradient descent optimization algorithm in pure Python — no ML libraries — demonstrating a ground-up understanding of how models learn.
- Validated convergence on a custom regression task; achieved ~94% cost reduction over 500 iterations, with annotated plots comparing learning rate effects.
- Modular design allows plug-in of any differentiable cost function; used as a learning reference by peers in class.

Python GUI Utility Suite | *Python, Tkinter, Pillow* [GitHub ↗](#)

- Built a desktop application with two production-ready tools: an Image Compression Tool and an Image Resizer — both with full GUI, file browser, and live preview.
- Implemented algorithmic logic for lossy compression and dimension scaling; achieved up to 70% file size reduction with minimal perceptible quality loss.
- Zero third-party ML dependencies; entire stack built on Python standard library and Pillow.

F1 Telemetry Analysis (In Progress) | *Python, FastF1, pandas, matplotlib* [GitHub ↗](#)

- Pulling lap-by-lap telemetry data via FastF1 API for the 2026 season; comparing throttle, braking, and speed traces across drivers at multiple circuits.
- Building an automated pipeline that generates race-comparison plots and highlights sector-by-sector performance deltas.
- Targeting a publishable analysis blog post on 3 races by end of semester.

EDUCATION

Jaypee University of Engineering and Technology (JUET) B.Tech in Computer Science and Engineering Currently 3rd Year	CGPA 7.7
Shri Gurukul Sr. Sec. School, Guna 12th (Senior Secondary)	85.6%
Shri Gurukul Sr. Sec. School, Guna 10th (Secondary)	93.4%

INTERESTS & LEARNING

Currently exploring: Applied ML for predictive analytics, introductory cybersecurity (CTF challenges), classical literature. Active on LinkedIn for professional networking and on GitHub for open-source contributions.